

Solutions to Selected Problems on HW#4

Note: In pretty much any mathematical problem beyond the level of high school algebra, there will be many possible solutions and many ways to elegantly write up the solution. So while your solution need not look exactly like what's here, it should be direct and clearly written.

41 (pg. 67):

Using the natural isomorphism between U_n and $\mathbb{Z}/n\mathbb{Z}$, we can find the primitive n^{th} roots of unity as follows: First observe that the generators for $\mathbb{Z}/12\mathbb{Z}$ are exactly the elements with a multiplicative inverse. These are $\{1, 5, 7, 11\}$.

Now, the isomorphism ϕ going from $\mathbb{Z}/n\mathbb{Z}$ to U_n is simply $\phi(x) := e^{\frac{2\pi i x}{12}}$. ($\phi(x) := e^{\frac{\pi i x}{6}}$, more concisely.) So the primitive 12^{th} roots of unity are then

$$\left\{ e^{\frac{\pi i}{6}}, e^{\frac{5\pi i}{6}}, e^{\frac{7\pi i}{6}}, e^{\frac{11\pi i}{6}} \right\}.$$

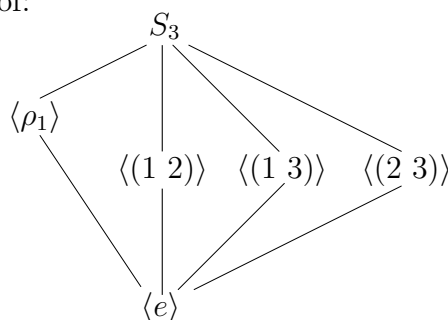
18 (pg. 84):

(a) As observed in class on Sep. 20, one can easily check by hand that $\rho_1^2 = \rho_2$, $\rho_2^2 = \rho_1$, and $\rho_1 \rho_2 = \rho_1^3 = \rho_2^3 = \tau^2 = e$. So we obtain

$$\begin{aligned} \langle \rho_1 \rangle &= \{e, \rho_1, \rho_2\} \\ \langle \rho_2 \rangle &= \{e, \rho_2, \rho_1\} \\ \langle \tau \rangle &= \{e, \tau\} \end{aligned}$$

In particular, $\langle \rho_1 \rangle = \langle \rho_2 \rangle$.

(b) First the diagram, then the proof:



Note that the containments (going upward) are all obvious, so the only thing left to do is prove that there the stated subgroups comprise *all* the subgroups of S_3 . This is most easily seen via Lagrange's Theorem, which tells us that any subgroup of S_3 has cardinality 1, 2, 3, or 6. There is exactly one subgroup of cardinality 1 — the trivial subgroup $\langle e \rangle$. There is also exactly one subgroup of cardinality 6 — the improper subgroup S_3 .

To account for the other subgroups, note that we proved earlier in class that all groups of order ≤ 3 are cyclic. (We did this via simple ad hoc arguments. Alternatively, this follows easily from Lagrange's Theorem.) Therefore, each subgroup of S_3 is generated by a single element. Enumerating all 6 elements of S_3 , we see that there are exactly 3 elements of order 2, each generating separate subgroups of order 2. Also, there are exactly 2 elements of order

3 and, from Part (a) above, we see that they both generate the same subgroup of order 3. So we are done. ■

10 (pg. 94):

Looking directly at the permutation $\begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 8 & 2 & 6 & 3 & 7 & 4 & 5 & 1 \end{bmatrix}$, we simply examine the orbits of the elements 1, 2, etc. in order, until all the elements of $\{1, \dots, 8\}$ appear in some cycle:

- $1 \mapsto 8 \mapsto 1$ yields the cycle $(1\ 8)$.
- $2 \mapsto 2$ yields the trivial cycle (or 1-cycle) (2) , which we may ignore later on.
- $3 \mapsto 6 \mapsto 4 \mapsto 3$ yields the transposition (or 2-cycle) $(3\ 6\ 4)$.
- $5 \mapsto 7 \mapsto 5$ yields the 3-cycle $(5\ 7)$.

At this point, all elements have been covered, and we obtain the decomposition

$$\boxed{\begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 8 & 2 & 6 & 3 & 7 & 4 & 5 & 1 \end{bmatrix} = (1\ 8)(3\ 6\ 4)(5\ 7)}.$$

To obtain a decomposition into transpositions, we merely apply the trick from lecture on Sep. 18 to each cycle: the transpositions can be left alone, and $(3\ 6\ 4) = (3\ 6)(3\ 4)$. So we obtain

$$\boxed{\begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 8 & 2 & 6 & 3 & 7 & 4 & 5 & 1 \end{bmatrix} = (1\ 8)(3\ 6)(3\ 4)(5\ 7)},$$

which is an even permutation.